

Institute of Assembly Technology and Industrial Robotics

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Master's Thesis

Sample Title

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Declaration

I hereby declare that I have written the present Master's Thesis independently and have not used any sources or aids other than those indicated. All parts of the thesis that are taken word for word or in essence from other sources are marked as such.

Regarding the use of so-called AI tools (ChatGPT, GitHub CoPilot, MS-Copilot, Google Gemini, etc.), I declare that they were used for the following purposes:

- Literature research
- Language, style improvement, or checking spelling and grammar
- Generation of text, e.g., creating a summary
- Generation of code
- Data analysis and/or visualization

I have critically evaluated all AI tools I have used, and the responsibility for the final results of the thesis lies entirely with me. All content derived from AI-generated suggestions has been checked by me for accuracy and relevance.

I further assure that the thesis has not been submitted to any examination authority in the same or similar form.

Place, Date

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Kurzfassung

Kurzfassung der Arbeit.

Abstract

Brief summary of the thesis.

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Formula and Symbol Directory

Symbol	Unit	Description
A_0	m^2	Initial contact area
a	m	Minor diameter of the ellipse
A	m^2	Area
B_0	$\frac{N}{s^2}$	Constant
B_1	$\frac{N}{s}$	Constant
B_2	N	Constant
B_3	Ns	Integration constant
B_4	Ns^2	Integration constant
b	m	Width
d	m	Major diameter of the ellipse

This page is a longtable. It automatically extends to the next pages if there are enough entries. The following website is helpful for this:

http://www.tablesgenerator.com/latex_tables

In the **File** tab, the **Paste table data** submenu can be used to convert an Excel table into LaTeX format.

1 General Information

Useful links:

- Online Table Generator
http://www.tablesgenerator.com/latex_tables
- Flowcharts and drawings easily online
<https://www.draw.io>
- Minipage Help
<http://www.namsu.de/Extra/befehle/Minipage.html>
- Online Formula Editor
<https://www.matheretter.de/tools/formeleditor/>
- Online Formula Editor with PDF Export
<https://www.latex4technics.com>

For linking a video, a QR code is helpful. Here it is shown as an example.



2 Fundamentals

2.1 Equations

Inclusion of equations using the *equation* command and referencing using the *eqref* command (see Equation (2.1)).

$$e^{i\pi} + 1 = 0 \quad (2.1)$$

An equation spread over multiple lines is exemplarily represented here.

$$\begin{aligned} \dot{\alpha} = \frac{1}{J_{zz}} \cdot \left[\frac{b}{2} \cdot \left(\int F_{R1} dt + \int F_{R4} dt - \left(\int F_{R2} + F_{R3} dt \right) \right) + \right. \\ \left. \frac{l}{4} \cdot \left(\int F_2 dt + \int F_3 dt - \left(\int F_1 dt + \int F_4 dt \right) \right) \right] + \alpha_0 \end{aligned} \quad (2.2)$$

A multiline listing of vectors is exemplarily shown here.

$$\vec{r}_{P1} = \begin{bmatrix} -b/2 \\ l/4 \\ h/4 \end{bmatrix} \quad \vec{r}_{P2} = \begin{bmatrix} -b/2 \\ -l/4 \\ h/4 \end{bmatrix} \quad \vec{r}_{P3} = \begin{bmatrix} b/2 \\ l/4 \\ h/4 \end{bmatrix} \quad \vec{r}_{P4} = \begin{bmatrix} b/2 \\ -l/4 \\ h/4 \end{bmatrix} \quad (2.3)$$

$$\vec{r}_{P5} = \begin{bmatrix} -b/2 \\ l/4 \\ -h/4 \end{bmatrix} \quad \vec{r}_{P6} = \begin{bmatrix} -b/2 \\ -l/4 \\ -h/4 \end{bmatrix} \quad \vec{r}_{P7} = \begin{bmatrix} b/2 \\ l/4 \\ -h/4 \end{bmatrix} \quad \vec{r}_{P8} = \begin{bmatrix} b/2 \\ -l/4 \\ -h/4 \end{bmatrix} \quad (2.4)$$

2.2 Bibliography

Citations are made using the *cite* command [1]. The cited literature must be present in `litvz.bib`. The Bib file can be created directly from Citavi, Mendeley, or a similar program.

2.3 Figure

Reference to a figure using the *ref* command (see Figure 2.1). Create vector graphics e.g., with Microsoft Visio, InkScape, Gimp, etc., export as PDF, and embed here. The *trim* command allows for cropping the graphic. Caption images directly in LaTeX (via *put*), allowing for consistent text style and special characters. The *caption* is used for the image caption.

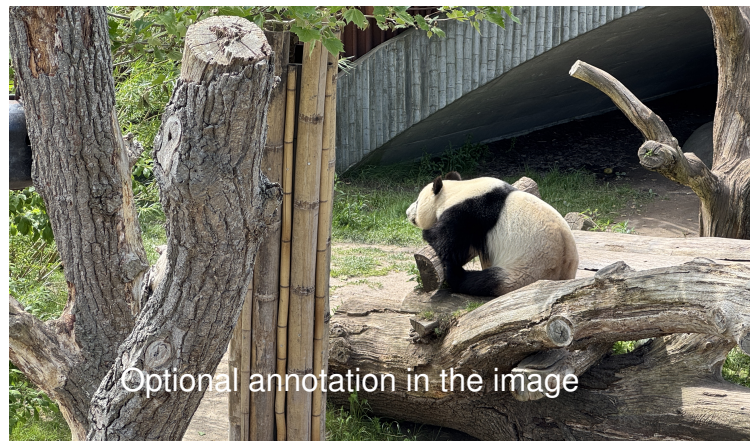


Figure 2.1: This is the image of a thoughtful panda (Image: Henrik Lurz)

To arrange multiple images or plots, the *subfigure* environment can be used (see 2.2). Here, multiple figures can be inserted within the (*figure*) environment. The arrangement depends greatly on the chosen width of the images.

2.4 Tables

An elegant caption for a table is described below.

Table 2.1: Measured values

Experiment \ Membrane	1	2	3	4	Angle
	[kPa]	[kPa]	[kPa]	[kPa]	[degrees]
1	30	0	30	0	-20



(a) Small panda (Image: Henrik Lurz)



(b) Large panda (Image: Henrik Lurz)



(c) Brave panda (Image: Henrik Lurz)



(d) Wannabe panda (Image: Henrik Lurz)

Figure 2.2: Even more pandas

2.5 Embedding Code Sections

Formatted code lines can be automatically embedded with the *lstlisting* package; the programming language used must be specified.

Programmbeispiel 1: Ein bisschen Matlab Code

```
1 %%  
2 % Minimalbeispiel  
3 clc  
4 clear  
5 % Festgesetzte Werte:  
6 p_0=0.4;      % bar  
7 a=0.03;       % m  
8 b=0.04;       % m  
9 m=2;          % kg  
10 t_E=1.3;      % s
```

```
11 t=0; % s
12 %% Berechnung der Konstanten
```


3 Bibliography

- [1] HESSE, Stefan: *Greifertechnik - Effektoren für Roboter und Automaten*. München : Hanser, 2011. – ISBN 978–3–446–42422–7

4 Appendix

Additional plots, data or code can be accommodated here.